

```
import pandas as pd
df = pd.read_csv(r"C:\Users\binarystudy\Projects\students_data.csv")
```

```
df
```

	student_id	name	age	gender	grade	math_score
0	100	jane smith	16.0	female	11	75.0
1	101	John Doe	16.0	Male	10th	74.0
2	102	Chris P.	NaN	MALE	10	NaN
3	103	jane smith	16.0	FEMALE	10	NaN
4	104	Sara O'Neil	16.0	male	11	NaN
5	105	Mike O'Reilly	16.0	Female	10	NaN
6	106	ali Khan	17.0	female	11	64.0
7	107	Sara O'Neil	17.0	female	12	NaN
8	108	Mike O'Reilly	16.0	Female	12	80.0
9	109	Robert Brown	17.0	female	12	NaN
10	110	lucy gray	17.0	male	11th	65.0
11	111	Simran Singh	16.0	FEMALE	11th	NaN
12	112	Patel R.	17.0	female	11	NaN
13	113	Patel R.	17.0	male	10	NaN
14	114	Ali Khan	17.0	male	12	NaN
15	115	Lucy gray	16.0	Female	12	65.0
16	116	Chris P.	NaN	Female	11	NaN
17	117	Ali Khan	18.0	male	11th	67.0
18	118	Simran Singh	17.0	Male	10	100.0
19	119	Patel R.	17.0	MALE	11th	73.0
20	120	Sara O'Neil	17.0	Male	10	NaN

21	121	John Doe	18.0	female	11	66.0
72						
22	122	Sara O'Neil	17.0	MALE	11th	75.0
NaN						
23	123	jane smith	17.0	Female	11th	NaN
missing						
24	124	John Doe	18.0	MALE	11th	NaN
NaN						
25	125	Mike O'Reilly	17.0	Male	12	94.0
80						
26	126	John Doe	NaN	Male	10th	64.0
missing						
27	127	Simran Singh	16.0	FEMALE	12	NaN
76						
28	128	Sara O'Neil	17.0	female	11	NaN
64						
29	129	Patel R.	17.0	male	11	NaN
64						
30	129	Patel R.	17.0	male	11	NaN
64						

	science_score	enrolled_date	remarks
0	66	2022-06-10	excellent
1	94	10-06-2022	GOOD
2	69	06/12/2022	needs improvement
3	62	10-06-2022	average
4	64	2022-06-10	GOOD
5	83	06/12/2022	needs improvement
6	75	06/12/2022	Good
7	62	2022/06/11	excellent
8	89	06/12/2022	poor
9	97	10-06-2022	needs improvement
10	100	06/12/2022	excellent
11	95	2022-06-10	average
12	89	2022-06-10	poor
13	98	06/12/2022	Average
14	67	2022/06/11	poor
15	94	06/12/2022	Average
16	72	2022/06/11	excellent
17	81	2022-06-10	GOOD
18	62	2022/06/11	average
19	90	06/12/2022	Good
20	89	2022-06-10	average
21	94	10-06-2022	Average
22	66	2022-06-10	good student
23	63	06/12/2022	excellent
24	91	06/12/2022	GOOD
25	63	10-06-2022	Average
26	67	06/12/2022	good student

27	80	06/12/2022	poor
28	89	2022-06-10	average
29	83	2022/06/11	Average
30	83	2022/06/11	Average

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 31 entries, 0 to 30
```

```
Data columns (total 10 columns):
```

#	Column	Non-Null Count	Dtype
0	student_id	31 non-null	int64
1	name	31 non-null	object
2	age	28 non-null	float64
3	gender	31 non-null	object
4	grade	31 non-null	object
5	math_score	13 non-null	float64
6	english_score	23 non-null	object
7	science_score	31 non-null	int64
8	enrolled_date	31 non-null	object
9	remarks	31 non-null	object

```
dtypes: float64(2), int64(2), object(6)
```

```
memory usage: 2.6+ KB
```

```
df.duplicated()
```

0	False
1	False
2	False
3	False
4	False
5	False
6	False
7	False
8	False
9	False
10	False
11	False
12	False
13	False
14	False
15	False
16	False
17	False
18	False
19	False
20	False
21	False
22	False

```
23     False
24     False
25     False
26     False
27     False
28     False
29     False
30     True
dtype: bool
```

```
df = df.drop_duplicates()
```

```
df
```

	student_id	name	age	gender	grade	math_score
0	100	jane smith	16.0	female	11	75.0
1	101	John Doe	16.0	Male	10th	74.0
2	102	Chris P.	NaN	MALE	10	NaN
3	103	jane smith	16.0	FEMALE	10	NaN
4	104	Sara O'Neil	16.0	male	11	NaN
5	105	Mike O'Reilly	16.0	Female	10	NaN
6	106	ali Khan	17.0	female	11	64.0
7	107	Sara O'Neil	17.0	female	12	NaN
8	108	Mike O'Reilly	16.0	Female	12	80.0
9	109	Robert Brown	17.0	female	12	NaN
10	110	lucy gray	17.0	male	11th	65.0
11	111	Simran Singh	16.0	FEMALE	11th	NaN
12	112	Patel R.	17.0	female	11	NaN
13	113	Patel R.	17.0	male	10	NaN
14	114	Ali Khan	17.0	male	12	NaN
15	115	Lucy gray	16.0	Female	12	65.0
16	116	Chris P.	NaN	Female	11	NaN

17	117	Ali Khan	18.0	male	11th	67.0
74						
18	118	Simran Singh	17.0	Male	10	100.0
74						
19	119	Patel R.	17.0	MALE	11th	73.0
NaN						
20	120	Sara O'Neil	17.0	Male	10	NaN
missing						
21	121	John Doe	18.0	female	11	66.0
72						
22	122	Sara O'Neil	17.0	MALE	11th	75.0
NaN						
23	123	jane smith	17.0	Female	11th	NaN
missing						
24	124	John Doe	18.0	MALE	11th	NaN
NaN						
25	125	Mike O'Reilly	17.0	Male	12	94.0
80						
26	126	John Doe	NaN	Male	10th	64.0
missing						
27	127	Simran Singh	16.0	FEMALE	12	NaN
76						
28	128	Sara O'Neil	17.0	female	11	NaN
64						
29	129	Patel R.	17.0	male	11	NaN
64						
	science_score	enrolled_date			remarks	
0	66	2022-06-10			excellent	
1	94	10-06-2022			GOOD	
2	69	06/12/2022	needs	improvement		
3	62	10-06-2022			average	
4	64	2022-06-10			GOOD	
5	83	06/12/2022	needs	improvement		
6	75	06/12/2022			Good	
7	62	2022/06/11			excellent	
8	89	06/12/2022			poor	
9	97	10-06-2022	needs	improvement		
10	100	06/12/2022			excellent	
11	95	2022-06-10			average	
12	89	2022-06-10			poor	
13	98	06/12/2022			Average	
14	67	2022/06/11			poor	
15	94	06/12/2022			Average	
16	72	2022/06/11			excellent	
17	81	2022-06-10			GOOD	
18	62	2022/06/11			average	
19	90	06/12/2022			Good	
20	89	2022-06-10			average	

21	94	10-06-2022	Average
22	66	2022-06-10	good student
23	63	06/12/2022	excellent
24	91	06/12/2022	GOOD
25	63	10-06-2022	Average
26	67	06/12/2022	good student
27	80	06/12/2022	poor
28	89	2022-06-10	average
29	83	2022/06/11	Average

```
df.isnull().sum()
```

```
student_id      0
name            0
age             3
gender          0
grade           0
math_score      17
english_score    8
science_score    0
enrolled_date    0
remarks         0
dtype: int64
```

```
df.loc[:, "name"] = df["name"].str.lower().str.title()
```

```
df
```

	student_id	name	age	gender	grade	math_score
english_score \						
0	100	Jane Smith	16.0	female	11	75.0
NaN						
1	101	John Doe	16.0	Male	10th	74.0
95						
2	102	Chris P.	NaN	MALE	10	NaN
missing						
3	103	Jane Smith	16.0	FEMALE	10	NaN
missing						
4	104	Sara O'Neil	16.0	male	11	NaN
96						
5	105	Mike O'Reilly	16.0	Female	10	NaN
NaN						
6	106	Ali Khan	17.0	female	11	64.0
NaN						
7	107	Sara O'Neil	17.0	female	12	NaN
63						
8	108	Mike O'Reilly	16.0	Female	12	80.0
missing						
9	109	Robert Brown	17.0	female	12	NaN
missing						

1067	110	Lucy Gray	17.0	male	11th	65.0
11missing	111	Simran Singh	16.0	FEMALE	11th	NaN
1287	112	Patel R.	17.0	female	11	NaN
13NaN	113	Patel R.	17.0	male	10	NaN
1491	114	Ali Khan	17.0	male	12	NaN
1591	115	Lucy Gray	16.0	Female	12	65.0
16NaN	116	Chris P.	NaN	Female	11	NaN
1774	117	Ali Khan	18.0	male	11th	67.0
1874	118	Simran Singh	17.0	Male	10	100.0
19NaN	119	Patel R.	17.0	MALE	11th	73.0
20missing	120	Sara O'Neil	17.0	Male	10	NaN
2172	121	John Doe	18.0	female	11	66.0
22NaN	122	Sara O'Neil	17.0	MALE	11th	75.0
23missing	123	Jane Smith	17.0	Female	11th	NaN
24NaN	124	John Doe	18.0	MALE	11th	NaN
2580	125	Mike O'Reilly	17.0	Male	12	94.0
26missing	126	John Doe	NaN	Male	10th	64.0
2776	127	Simran Singh	16.0	FEMALE	12	NaN
2864	128	Sara O'Neil	17.0	female	11	NaN
2964	129	Patel R.	17.0	male	11	NaN
	science_score	enrolled_date		remarks		
0	66	2022-06-10		excellent		
1	94	10-06-2022		GOOD		
2	69	06/12/2022	needs	improvement		
3	62	10-06-2022		average		
4	64	2022-06-10		GOOD		
5	83	06/12/2022	needs	improvement		
6	75	06/12/2022		Good		

7	62	2022/06/11	excellent
8	89	06/12/2022	poor
9	97	10-06-2022	needs improvement
10	100	06/12/2022	excellent
11	95	2022-06-10	average
12	89	2022-06-10	poor
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25	63	10-06-2022	Average
26	67	06/12/2022	good student
27	80	06/12/2022	poor
28	89	2022-06-10	average
29	83	2022/06/11	Average

```
df["age"].unique()
```

```
array([16., nan, 17., 18.])
```

```
age_mean = df["age"].mean().round(0)
```

```
df.loc[:, "age"] = df["age"].fillna(age_mean)
```

```
df
```

	student_id	name	age	gender	grade	math_score
0	100	Jane Smith	16.0	female	11	75.0
1	101	John Doe	16.0	Male	10th	74.0
2	102	Chris P.	17.0	MALE	10	NaN
3	103	Jane Smith	16.0	FEMALE	10	NaN
4	104	Sara O'Neil	16.0	male	11	NaN
5	105	Mike O'Reilly	16.0	Female	10	NaN
6	106	Ali Khan	17.0	female	11	64.0
7	107	Sara O'Neil	17.0	female	12	NaN

63							
8	108	Mike O'Reilly	16.0	Female	12	80.0	
missing							
9	109	Robert Brown	17.0	female	12	NaN	
missing							
10	110	Lucy Gray	17.0	male	11th	65.0	
67							
11	111	Simran Singh	16.0	FEMALE	11th	NaN	
missing							
12	112	Patel R.	17.0	female	11	NaN	
87							
13	113	Patel R.	17.0	male	10	NaN	
NaN							
14	114	Ali Khan	17.0	male	12	NaN	
91							
15	115	Lucy Gray	16.0	Female	12	65.0	
91							
16	116	Chris P.	17.0	Female	11	NaN	
NaN							
17	117	Ali Khan	18.0	male	11th	67.0	
74							
18	118	Simran Singh	17.0	Male	10	100.0	
74							
19	119	Patel R.	17.0	MALE	11th	73.0	
NaN							
20	120	Sara O'Neil	17.0	Male	10	NaN	
missing							
21	121	John Doe	18.0	female	11	66.0	
72							
22	122	Sara O'Neil	17.0	MALE	11th	75.0	
NaN							
23	123	Jane Smith	17.0	Female	11th	NaN	
missing							
24	124	John Doe	18.0	MALE	11th	NaN	
NaN							
25	125	Mike O'Reilly	17.0	Male	12	94.0	
80							
26	126	John Doe	17.0	Male	10th	64.0	
missing							
27	127	Simran Singh	16.0	FEMALE	12	NaN	
76							
28	128	Sara O'Neil	17.0	female	11	NaN	
64							
29	129	Patel R.	17.0	male	11	NaN	
64							
	science_score	enrolled_date	remarks				
0	66	2022-06-10	excellent				
1	94	10-06-2022	GOOD				

2	69	06/12/2022	needs improvement
3	62	10-06-2022	average
4	64	2022-06-10	GOOD
5	83	06/12/2022	needs improvement
6	75	06/12/2022	Good
7	62	2022/06/11	excellent
8	89	06/12/2022	poor
9	97	10-06-2022	needs improvement
10	100	06/12/2022	excellent
11	95	2022-06-10	average
12	89	2022-06-10	poor
13	98	06/12/2022	Average
14	67	2022/06/11	poor
15	94	06/12/2022	Average
16	72	2022/06/11	excellent
17	81	2022-06-10	GOOD
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19	90	06/12/2022	Good
20	89	2022-06-10	average
21	94	10-06-2022	Average
22	66	2022-06-10	good student
23	63	06/12/2022	excellent
24	91	06/12/2022	GOOD
25	63	10-06-2022	Average
26	67	06/12/2022	good student
27	80	06/12/2022	poor
28	89	2022-06-10	average
29	83	2022/06/11	Average

```
df["gender"].unique()
```

```
array(['female', 'Male', 'MALE', 'FEMALE', 'male', 'Female'],
      dtype=object)
```

```
df.loc[:, "gender"] = df["gender"].str.lower().str.title()
```

```
df
```

	student_id	name	age	gender	grade	math_score
0	100	Jane Smith	16.0	Female	11	75.0
1	101	John Doe	16.0	Male	10th	74.0
2	102	Chris P.	17.0	Male	10	NaN
3	103	Jane Smith	16.0	Female	10	NaN
4	104	Sara O'Neil	16.0	Male	11	NaN

5	105	Mike O'Reilly	16.0	Female	10	NaN
NaN						
6	106	Ali Khan	17.0	Female	11	64.0
NaN						
7	107	Sara O'Neil	17.0	Female	12	NaN
63						
8	108	Mike O'Reilly	16.0	Female	12	80.0
missing						
9	109	Robert Brown	17.0	Female	12	NaN
missing						
10	110	Lucy Gray	17.0	Male	11th	65.0
67						
11	111	Simran Singh	16.0	Female	11th	NaN
missing						
12	112	Patel R.	17.0	Female	11	NaN
87						
13	113	Patel R.	17.0	Male	10	NaN
NaN						
14	114	Ali Khan	17.0	Male	12	NaN
91						
15	115	Lucy Gray	16.0	Female	12	65.0
91						
16	116	Chris P.	17.0	Female	11	NaN
NaN						
17	117	Ali Khan	18.0	Male	11th	67.0
74						
18	118	Simran Singh	17.0	Male	10	100.0
74						
19	119	Patel R.	17.0	Male	11th	73.0
NaN						
20	120	Sara O'Neil	17.0	Male	10	NaN
missing						
21	121	John Doe	18.0	Female	11	66.0
72						
22	122	Sara O'Neil	17.0	Male	11th	75.0
NaN						
23	123	Jane Smith	17.0	Female	11th	NaN
missing						
24	124	John Doe	18.0	Male	11th	NaN
NaN						
25	125	Mike O'Reilly	17.0	Male	12	94.0
80						
26	126	John Doe	17.0	Male	10th	64.0
missing						
27	127	Simran Singh	16.0	Female	12	NaN
76						
28	128	Sara O'Neil	17.0	Female	11	NaN
64						
29	129	Patel R.	17.0	Male	11	NaN

64

	science_score	enrolled_date	remarks
0	66	2022-06-10	excellent
1	94	10-06-2022	GOOD
2	69	06/12/2022	needs improvement
3	62	10-06-2022	average
4	64	2022-06-10	GOOD
5	83	06/12/2022	needs improvement
6	75	06/12/2022	Good
7	62	2022/06/11	excellent
8	89	06/12/2022	poor
9	97	10-06-2022	needs improvement
10	100	06/12/2022	excellent
11	95	2022-06-10	average
12	89	2022-06-10	poor
13	98	06/12/2022	Average
14	67	2022/06/11	poor
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16	72	2022/06/11	excellent
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19	90	06/12/2022	Good
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23	63	06/12/2022	excellent
24	91	06/12/2022	GOOD
25	63	10-06-2022	Average
26	67	06/12/2022	good student
27	80	06/12/2022	poor
28	89	2022-06-10	average
29	83	2022/06/11	Average

```
df["grade"]
```

0	11
1	10th
2	10
3	10
4	11
5	10
6	11
7	12
8	12
9	12
10	11th
11	11th
12	11
13	10

```
14      12
15      12
16      11
17     11th
18      10
19     11th
20      10
21      11
22     11th
23     11th
24     11th
25      12
26     10th
27      12
28      11
29      11
```

Name: grade, dtype: object

```
df["grade"].unique()
```

```
array(['11', '10th', '10', '12', '11th'], dtype=object)
```

```
df.loc[:, "grade"] = df["grade"].replace({'10th':'10', '11th':'11'})
```

```
df
```

	student_id	name	age	gender	grade	math_score
0	100	Jane Smith	16.0	Female	11	75.0
1	101	John Doe	16.0	Male	10	74.0
2	102	Chris P.	17.0	Male	10	NaN
3	103	Jane Smith	16.0	Female	10	NaN
4	104	Sara O'Neil	16.0	Male	11	NaN
5	105	Mike O'Reilly	16.0	Female	10	NaN
6	106	Ali Khan	17.0	Female	11	64.0
7	107	Sara O'Neil	17.0	Female	12	NaN
8	108	Mike O'Reilly	16.0	Female	12	80.0
9	109	Robert Brown	17.0	Female	12	NaN
10	110	Lucy Gray	17.0	Male	11	65.0

11	111	Simran Singh	16.0	Female	11	NaN
missing						
12	112	Patel R.	17.0	Female	11	NaN
87						
13	113	Patel R.	17.0	Male	10	NaN
NaN						
14	114	Ali Khan	17.0	Male	12	NaN
91						
15	115	Lucy Gray	16.0	Female	12	65.0
91						
16	116	Chris P.	17.0	Female	11	NaN
NaN						
17	117	Ali Khan	18.0	Male	11	67.0
74						
18	118	Simran Singh	17.0	Male	10	100.0
74						
19	119	Patel R.	17.0	Male	11	73.0
NaN						
20	120	Sara O'Neil	17.0	Male	10	NaN
missing						
21	121	John Doe	18.0	Female	11	66.0
72						
22	122	Sara O'Neil	17.0	Male	11	75.0
NaN						
23	123	Jane Smith	17.0	Female	11	NaN
missing						
24	124	John Doe	18.0	Male	11	NaN
NaN						
25	125	Mike O'Reilly	17.0	Male	12	94.0
80						
26	126	John Doe	17.0	Male	10	64.0
missing						
27	127	Simran Singh	16.0	Female	12	NaN
76						
28	128	Sara O'Neil	17.0	Female	11	NaN
64						
29	129	Patel R.	17.0	Male	11	NaN
64						
	science_score	enrolled_date	remarks			
0	66	2022-06-10	excellent			
1	94	10-06-2022	GOOD			
2	69	06/12/2022	needs improvement			
3	62	10-06-2022	average			
4	64	2022-06-10	GOOD			
5	83	06/12/2022	needs improvement			
6	75	06/12/2022	Good			
7	62	2022/06/11	excellent			
8	89	06/12/2022	poor			

9	97	10-06-2022	needs improvement
10	100	06/12/2022	excellent
11	95	2022-06-10	average
12	89	2022-06-10	poor
13	98	06/12/2022	Average
14	67	2022/06/11	poor
15	94	06/12/2022	Average
16	72	2022/06/11	excellent
17	81	2022-06-10	GOOD
18	62	2022/06/11	average
19	90	06/12/2022	Good
20	89	2022-06-10	average
21	94	10-06-2022	Average
22	66	2022-06-10	good student
23	63	06/12/2022	excellent
24	91	06/12/2022	GOOD
25	63	10-06-2022	Average
26	67	06/12/2022	good student
27	80	06/12/2022	poor
28	89	2022-06-10	average
29	83	2022/06/11	Average

```
math_mean = df["math_score"].mean()
```

```
df.loc[:, "math_score"] = df["math_score"].fillna(math_mean)
```

```
df
```

	student_id	name	age	gender	grade	math_score
english_score \						
0	100	Jane Smith	16.0	Female	11	75.0
NaN						
1	101	John Doe	16.0	Male	10	74.0
95						
2	102	Chris P.	17.0	Male	10	74.0
missing						
3	103	Jane Smith	16.0	Female	10	74.0
missing						
4	104	Sara O'Neil	16.0	Male	11	74.0
96						
5	105	Mike O'Reilly	16.0	Female	10	74.0
NaN						
6	106	Ali Khan	17.0	Female	11	64.0
NaN						
7	107	Sara O'Neil	17.0	Female	12	74.0
63						
8	108	Mike O'Reilly	16.0	Female	12	80.0
missing						
9	109	Robert Brown	17.0	Female	12	74.0
missing						

10	110	Lucy Gray	17.0	Male	11	65.0
67						
11	111	Simran Singh	16.0	Female	11	74.0
missing						
12	112	Patel R.	17.0	Female	11	74.0
87						
13	113	Patel R.	17.0	Male	10	74.0
NaN						
14	114	Ali Khan	17.0	Male	12	74.0
91						
15	115	Lucy Gray	16.0	Female	12	65.0
91						
16	116	Chris P.	17.0	Female	11	74.0
NaN						
17	117	Ali Khan	18.0	Male	11	67.0
74						
18	118	Simran Singh	17.0	Male	10	100.0
74						
19	119	Patel R.	17.0	Male	11	73.0
NaN						
20	120	Sara O'Neil	17.0	Male	10	74.0
missing						
21	121	John Doe	18.0	Female	11	66.0
72						
22	122	Sara O'Neil	17.0	Male	11	75.0
NaN						
23	123	Jane Smith	17.0	Female	11	74.0
missing						
24	124	John Doe	18.0	Male	11	74.0
NaN						
25	125	Mike O'Reilly	17.0	Male	12	94.0
80						
26	126	John Doe	17.0	Male	10	64.0
missing						
27	127	Simran Singh	16.0	Female	12	74.0
76						
28	128	Sara O'Neil	17.0	Female	11	74.0
64						
29	129	Patel R.	17.0	Male	11	74.0
64						
	science_score	enrolled_date	remarks			
0	66	2022-06-10	excellent			
1	94	10-06-2022	GOOD			
2	69	06/12/2022	needs improvement			
3	62	10-06-2022	average			
4	64	2022-06-10	GOOD			
5	83	06/12/2022	needs improvement			
6	75	06/12/2022	Good			

7	62	2022/06/11	excellent
8	89	06/12/2022	poor
9	97	10-06-2022	needs improvement
10	100	06/12/2022	excellent
11	95	2022-06-10	average
12	89	2022-06-10	poor
13	98	06/12/2022	Average
14	67	2022/06/11	poor
15	94	06/12/2022	Average
16	72	2022/06/11	excellent
17	81	2022-06-10	GOOD
18	62	2022/06/11	average
19	90	06/12/2022	Good
20	89	2022-06-10	average
21	94	10-06-2022	Average
22	66	2022-06-10	good student
23	63	06/12/2022	excellent
24	91	06/12/2022	GOOD
25	63	10-06-2022	Average
26	67	06/12/2022	good student
27	80	06/12/2022	poor
28	89	2022-06-10	average
29	83	2022/06/11	Average

```
df["english_score"]
```

0	NaN
1	95
2	missing
3	missing
4	96
5	NaN
6	NaN
7	63
8	missing
9	missing
10	67
11	missing
12	87
13	NaN
14	91
15	91
16	NaN
17	74
18	74
19	NaN
20	missing
21	72
22	NaN
23	missing

```
24      NaN
25      80
26    missing
27      76
28      64
29      64
```

```
Name: english_score, dtype: object
```

```
df.loc[:, "english_score"] = pd.to_numeric(df["english_score"],
errors='coerce')
```

```
df
```

	student_id	name	age	gender	grade	math_score
0	100	Jane Smith	16.0	Female	11	75.0
1	101	John Doe	16.0	Male	10	74.0
2	102	Chris P.	17.0	Male	10	74.0
3	103	Jane Smith	16.0	Female	10	74.0
4	104	Sara O'Neil	16.0	Male	11	74.0
5	105	Mike O'Reilly	16.0	Female	10	74.0
6	106	Ali Khan	17.0	Female	11	64.0
7	107	Sara O'Neil	17.0	Female	12	74.0
8	108	Mike O'Reilly	16.0	Female	12	80.0
9	109	Robert Brown	17.0	Female	12	74.0
10	110	Lucy Gray	17.0	Male	11	65.0
11	111	Simran Singh	16.0	Female	11	74.0
12	112	Patel R.	17.0	Female	11	74.0
13	113	Patel R.	17.0	Male	10	74.0
14	114	Ali Khan	17.0	Male	12	74.0
15	115	Lucy Gray	16.0	Female	12	65.0
16	116	Chris P.	17.0	Female	11	74.0
17	117	Ali Khan	18.0	Male	11	67.0

74.0						
18	118	Simran Singh	17.0	Male	10	100.0
74.0						
19	119	Patel R.	17.0	Male	11	73.0
NaN						
20	120	Sara O'Neil	17.0	Male	10	74.0
NaN						
21	121	John Doe	18.0	Female	11	66.0
72.0						
22	122	Sara O'Neil	17.0	Male	11	75.0
NaN						
23	123	Jane Smith	17.0	Female	11	74.0
NaN						
24	124	John Doe	18.0	Male	11	74.0
NaN						
25	125	Mike O'Reilly	17.0	Male	12	94.0
80.0						
26	126	John Doe	17.0	Male	10	64.0
NaN						
27	127	Simran Singh	16.0	Female	12	74.0
76.0						
28	128	Sara O'Neil	17.0	Female	11	74.0
64.0						
29	129	Patel R.	17.0	Male	11	74.0
64.0						

	science_score	enrolled_date	remarks
0	66	2022-06-10	excellent
1	94	10-06-2022	GOOD
2	69	06/12/2022	needs improvement
3	62	10-06-2022	average
4	64	2022-06-10	GOOD
5	83	06/12/2022	needs improvement
6	75	06/12/2022	Good
7	62	2022/06/11	excellent
8	89	06/12/2022	poor
9	97	10-06-2022	needs improvement
10	100	06/12/2022	excellent
11	95	2022-06-10	average
12	89	2022-06-10	poor
13	98	06/12/2022	Average
14	67	2022/06/11	poor
15	94	06/12/2022	Average
16	72	2022/06/11	excellent
17	81	2022-06-10	GOOD
18	62	2022/06/11	average
19	90	06/12/2022	Good
20	89	2022-06-10	average
21	94	10-06-2022	Average

22	66	2022-06-10	good student
23	63	06/12/2022	excellent
24	91	06/12/2022	GOOD
25	63	10-06-2022	Average
26	67	06/12/2022	good student
27	80	06/12/2022	poor
28	89	2022-06-10	average
29	83	2022/06/11	Average

```
df["enrolled_date"].unique()
```

```
array(['2022-06-10', '10-06-2022', '06/12/2022', '2022/06/11'],
      dtype=object)
```

```
## yyyy-mm-dd
```

```
dict_date={
    '10-06-2022': '2022-06-10',
    '06/12/2022': '2022-06-12',
    '2022/06/11': '2022-06-11'
}
```

```
df.loc[:, "enrolled_date"] = df["enrolled_date"].replace(dict_date)
```

```
df
```

	student_id	name	age	gender	grade	math_score
english_score \						
0	100	Jane Smith	16.0	Female	11	75.0
NaN						
1	101	John Doe	16.0	Male	10	74.0
95.0						
2	102	Chris P.	17.0	Male	10	74.0
NaN						
3	103	Jane Smith	16.0	Female	10	74.0
NaN						
4	104	Sara O'Neil	16.0	Male	11	74.0
96.0						
5	105	Mike O'Reilly	16.0	Female	10	74.0
NaN						
6	106	Ali Khan	17.0	Female	11	64.0
NaN						
7	107	Sara O'Neil	17.0	Female	12	74.0
63.0						
8	108	Mike O'Reilly	16.0	Female	12	80.0
NaN						
9	109	Robert Brown	17.0	Female	12	74.0
NaN						
10	110	Lucy Gray	17.0	Male	11	65.0
67.0						
11	111	Simran Singh	16.0	Female	11	74.0

NaN						
12	112	Patel R.	17.0	Female	11	74.0
87.0						
13	113	Patel R.	17.0	Male	10	74.0
NaN						
14	114	Ali Khan	17.0	Male	12	74.0
91.0						
15	115	Lucy Gray	16.0	Female	12	65.0
91.0						
16	116	Chris P.	17.0	Female	11	74.0
NaN						
17	117	Ali Khan	18.0	Male	11	67.0
74.0						
18	118	Simran Singh	17.0	Male	10	100.0
74.0						
19	119	Patel R.	17.0	Male	11	73.0
NaN						
20	120	Sara O'Neil	17.0	Male	10	74.0
NaN						
21	121	John Doe	18.0	Female	11	66.0
72.0						
22	122	Sara O'Neil	17.0	Male	11	75.0
NaN						
23	123	Jane Smith	17.0	Female	11	74.0
NaN						
24	124	John Doe	18.0	Male	11	74.0
NaN						
25	125	Mike O'Reilly	17.0	Male	12	94.0
80.0						
26	126	John Doe	17.0	Male	10	64.0
NaN						
27	127	Simran Singh	16.0	Female	12	74.0
76.0						
28	128	Sara O'Neil	17.0	Female	11	74.0
64.0						
29	129	Patel R.	17.0	Male	11	74.0
64.0						

	science_score	enrolled_date	remarks
0	66	2022-06-10	excellent
1	94	2022-06-10	GOOD
2	69	2022-06-12	needs improvement
3	62	2022-06-10	average
4	64	2022-06-10	GOOD
5	83	2022-06-12	needs improvement
6	75	2022-06-12	Good
7	62	2022-06-11	excellent
8	89	2022-06-12	poor
9	97	2022-06-10	needs improvement

10	100	2022-06-12	excellent
11	95	2022-06-10	average
12	89	2022-06-10	poor
13	98	2022-06-12	Average
14	67	2022-06-11	poor
15	94	2022-06-12	Average
16	72	2022-06-11	excellent
17	81	2022-06-10	GOOD
18	62	2022-06-11	average
19	90	2022-06-12	Good
20	89	2022-06-10	average
21	94	2022-06-10	Average
22	66	2022-06-10	good student
23	63	2022-06-12	excellent
24	91	2022-06-12	GOOD
25	63	2022-06-10	Average
26	67	2022-06-12	good student
27	80	2022-06-12	poor
28	89	2022-06-10	average
29	83	2022-06-11	Average

```
df["enrolled_date"] = pd.to_datetime(df["enrolled_date"])
```

C:\Users\91851\AppData\Local\Temp\ipykernel_7604\1499738188.py:1:

SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation:

https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
df["enrolled_date"] = pd.to_datetime(df["enrolled_date"])
```

df

	student_id	name	age	gender	grade	math_score
0	100	Jane Smith	16.0	Female	11	75.0
1	101	John Doe	16.0	Male	10	74.0
2	102	Chris P.	17.0	Male	10	74.0
3	103	Jane Smith	16.0	Female	10	74.0
4	104	Sara O'Neil	16.0	Male	11	74.0
5	105	Mike O'Reilly	16.0	Female	10	74.0
6	106	Ali Khan	17.0	Female	11	64.0

NaN						
7	107	Sara O'Neil	17.0	Female	12	74.0
63.0						
8	108	Mike O'Reilly	16.0	Female	12	80.0
NaN						
9	109	Robert Brown	17.0	Female	12	74.0
NaN						
10	110	Lucy Gray	17.0	Male	11	65.0
67.0						
11	111	Simran Singh	16.0	Female	11	74.0
NaN						
12	112	Patel R.	17.0	Female	11	74.0
87.0						
13	113	Patel R.	17.0	Male	10	74.0
NaN						
14	114	Ali Khan	17.0	Male	12	74.0
91.0						
15	115	Lucy Gray	16.0	Female	12	65.0
91.0						
16	116	Chris P.	17.0	Female	11	74.0
NaN						
17	117	Ali Khan	18.0	Male	11	67.0
74.0						
18	118	Simran Singh	17.0	Male	10	100.0
74.0						
19	119	Patel R.	17.0	Male	11	73.0
NaN						
20	120	Sara O'Neil	17.0	Male	10	74.0
NaN						
21	121	John Doe	18.0	Female	11	66.0
72.0						
22	122	Sara O'Neil	17.0	Male	11	75.0
NaN						
23	123	Jane Smith	17.0	Female	11	74.0
NaN						
24	124	John Doe	18.0	Male	11	74.0
NaN						
25	125	Mike O'Reilly	17.0	Male	12	94.0
80.0						
26	126	John Doe	17.0	Male	10	64.0
NaN						
27	127	Simran Singh	16.0	Female	12	74.0
76.0						
28	128	Sara O'Neil	17.0	Female	11	74.0
64.0						
29	129	Patel R.	17.0	Male	11	74.0
64.0						
science_score enrolled_date			remarks			

0	66	2022-06-10	excellent
1	94	2022-06-10	GOOD
2	69	2022-06-12	needs improvement
3	62	2022-06-10	average
4	64	2022-06-10	GOOD
5	83	2022-06-12	needs improvement
6	75	2022-06-12	Good
7	62	2022-06-11	excellent
8	89	2022-06-12	poor
9	97	2022-06-10	needs improvement
10	100	2022-06-12	excellent
11	95	2022-06-10	average
12	89	2022-06-10	poor
13	98	2022-06-12	Average
14	67	2022-06-11	poor
15	94	2022-06-12	Average
16	72	2022-06-11	excellent
17	81	2022-06-10	GOOD
18	62	2022-06-11	average
19	90	2022-06-12	Good
20	89	2022-06-10	average
21	94	2022-06-10	Average
22	66	2022-06-10	good student
23	63	2022-06-12	excellent
24	91	2022-06-12	GOOD
25	63	2022-06-10	Average
26	67	2022-06-12	good student
27	80	2022-06-12	poor
28	89	2022-06-10	average
29	83	2022-06-11	Average

```
df["enrolled_date"]
```

0	2022-06-10
1	2022-06-10
2	2022-06-12
3	2022-06-10
4	2022-06-10
5	2022-06-12
6	2022-06-12
7	2022-06-11
8	2022-06-12
9	2022-06-10
10	2022-06-12
11	2022-06-10
12	2022-06-10
13	2022-06-12
14	2022-06-11
15	2022-06-12
16	2022-06-11


```

17    2022-06-10
18    2022-06-11
19    2022-06-12
20    2022-06-10
21    2022-06-10
22    2022-06-10
23    2022-06-12
24    2022-06-12
25    2022-06-10
26    2022-06-12
27    2022-06-12
28    2022-06-10
29    2022-06-11
Name: enrolled_date, dtype: datetime64[ns]

df["remarks"].unique()

array(['excellent', 'GOOD', 'needs improvement', 'average', 'Good',
      'poor', 'Average', 'good student'], dtype=object)

dict_remark = {
    'excellent': 'Excellent',
    'GOOD': 'Good',
    'needs improvement': 'Needs Improvement',
    'average': 'Average',
    'poor': 'Poor',
    'good student': 'Good'
}

df["remarks"].replace(dict_remark).unique()

array(['Excellent', 'Good', 'Needs Improvement', 'Average', 'Poor'],
      dtype=object)

df.loc[:, "remarks"] = df["remarks"].replace(dict_remark)

df

```

	student_id	name	age	gender	grade	math_score
0	100	Jane Smith	16.0	Female	11	75.0
1	101	John Doe	16.0	Male	10	74.0
2	102	Chris P.	17.0	Male	10	74.0
3	103	Jane Smith	16.0	Female	10	74.0
4	104	Sara O'Neil	16.0	Male	11	74.0
5	105	Mike O'Reilly	16.0	Female	10	74.0

NaN						
6	106	Ali Khan	17.0	Female	11	64.0
NaN						
7	107	Sara O'Neil	17.0	Female	12	74.0
63.0						
8	108	Mike O'Reilly	16.0	Female	12	80.0
NaN						
9	109	Robert Brown	17.0	Female	12	74.0
NaN						
10	110	Lucy Gray	17.0	Male	11	65.0
67.0						
11	111	Simran Singh	16.0	Female	11	74.0
NaN						
12	112	Patel R.	17.0	Female	11	74.0
87.0						
13	113	Patel R.	17.0	Male	10	74.0
NaN						
14	114	Ali Khan	17.0	Male	12	74.0
91.0						
15	115	Lucy Gray	16.0	Female	12	65.0
91.0						
16	116	Chris P.	17.0	Female	11	74.0
NaN						
17	117	Ali Khan	18.0	Male	11	67.0
74.0						
18	118	Simran Singh	17.0	Male	10	100.0
74.0						
19	119	Patel R.	17.0	Male	11	73.0
NaN						
20	120	Sara O'Neil	17.0	Male	10	74.0
NaN						
21	121	John Doe	18.0	Female	11	66.0
72.0						
22	122	Sara O'Neil	17.0	Male	11	75.0
NaN						
23	123	Jane Smith	17.0	Female	11	74.0
NaN						
24	124	John Doe	18.0	Male	11	74.0
NaN						
25	125	Mike O'Reilly	17.0	Male	12	94.0
80.0						
26	126	John Doe	17.0	Male	10	64.0
NaN						
27	127	Simran Singh	16.0	Female	12	74.0
76.0						
28	128	Sara O'Neil	17.0	Female	11	74.0
64.0						
29	129	Patel R.	17.0	Male	11	74.0
64.0						

	science_score	enrolled_date	remarks
0	66	2022-06-10	Excellent
1	94	2022-06-10	Good
2	69	2022-06-12	Needs Improvement
3	62	2022-06-10	Average
4	64	2022-06-10	Good
5	83	2022-06-12	Needs Improvement
6	75	2022-06-12	Good
7	62	2022-06-11	Excellent
8	89	2022-06-12	Poor
9	97	2022-06-10	Needs Improvement
10	100	2022-06-12	Excellent
11	95	2022-06-10	Average
12	89	2022-06-10	Poor
13	98	2022-06-12	Average
14	67	2022-06-11	Poor
15	94	2022-06-12	Average
16	72	2022-06-11	Excellent
17	81	2022-06-10	Good
18	62	2022-06-11	Average
19	90	2022-06-12	Good
20	89	2022-06-10	Average
21	94	2022-06-10	Average
22	66	2022-06-10	Good
23	63	2022-06-12	Excellent
24	91	2022-06-12	Good
25	63	2022-06-10	Average
26	67	2022-06-12	Good
27	80	2022-06-12	Poor
28	89	2022-06-10	Average
29	83	2022-06-11	Average